ALYA RAHMATILLAH MACHMUD

Informatics Engineering Student · Full-Stack Developer · UI/UX Designer · Data & AI Enthusiast

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SUMMARY

Informatics Engineering student at Institut Teknologi Sepuluh Nopember (ITS) with a strong academic background in software engineering. Passionate about building full-stack web applications, designing intuitive UI/UX, and applying data analysis and machine learning to real-world problems. An endless learner committed to continuous growth and delivering high-quality solutions in collaborative, dynamic environments.

EDUCATION

Bachelor of Degree in Informatics Engineering

Institut Teknologi Sepuluh Nopember (ITS) — Surabaya | Aug 2023 – Jul 2026 (Expected)

- Awardee of Program Beasiswa Santri Berprestasi (PBSB) from Kementrian Agama and LPDP 2023–2027.

WORK EXPERIENCE

Full-Stack Web Developer Intern

PT. Bank Aceh Syariah Head Office — Banda Aceh | Jan 2026 – Present

- Engineered a scalable full-stack web application for internal data tracking, designed to handle long-term operations for thousands of active users.
- Designed end-to-end system architecture covering frontend interfaces, backend logic, and a highly structured database.
- Executed secure server integrations directly with Bank Aceh's internal servers while maintaining strict data integrity.

Web Developer Intern

ITS Library (LetsgoLibrary) — Surabaya | Sep 2025 – Present

- Developing an internal repository web application for the ITS Library using Laravel, with full frontend and backend responsibility.
- Designing UI/UX flows and efficient database structures for library collection management.
- Integrating the system with the MyITS portal for centralized user access.

Teaching Assistant — Fundamental Programming

Institut Teknologi Sepuluh Nopember | Nov 2024 – May 2025

- Conducted private tutoring for C and C++, covering core syntax, control structures, and object-oriented concepts.
- Assisted the lecturer in creating technical demos and exercises for loops, arrays, and functions.

ORGANIZATION & VOLUNTEER EXPERIENCE

Department of Entrepreneurship Development

Himpunan Mahasiswa Teknik Informatika (HMTC) ITS — Surabaya | Feb 2024 – Present

- Managed the organization's largest financial project (Official Association Jacket), handling budgeting, procurement, and distribution for hundreds of students.
- Achieved financial targets by designing and marketing creative merchandise while maintaining accurate departmental accounting records.

Head of Finance Unit

Ramadhan Di Kampus 2026 — Surabaya | Jun 2025 – Present

- Led a 38-member financial department overseeing Sponsorship, Infaq, and Fundraising divisions.
- Strategized and executed a comprehensive fundraising plan targeting IDR 700 million through corporate sponsorships and philanthropic donations.
- Managed end-to-end financial flow from revenue generation to budget allocation, ensuring liquidity and successful event execution.

Inspiring Tutor

Gerakan Mengajar Desa — East Java | Oct 2024 – Mar 2025

- Educated rural elementary students on basic numeracy and character building as part of a national-scale social initiative.
- Contributed to communal projects and preserved cultural heritage through traditional dance performances.

PROJECTS

Ringkaskeun — AI Document Summarizer

Python · Flask · NLP · AI Integration

AI-powered web platform for summarizing uploaded documents with features including document history tracking, auto-generated flashcards, and Q&A; sessions derived from the uploaded content. Integrated directly into the Senopati ITS ecosystem.

LetsGo — ITS Library Repository System

Laravel · PHP · JavaScript · MySQL · Apache2

Web-based information system for monitoring and preserving the book collection at ITS Library. Features include tracking status (document verification history from data entry to scanning), collection data management, role-based user management (Super Admin vs Admin), and transparent preservation activity reporting.

https://librigo.its.ac.id/letsgo/

SEVIMA AI Content Creator

Next.js · TypeScript · CSS Modules · Golang

Full-fledged AI-powered platform for automated marketing video content creation. Frontend built with Next.js (TypeScript & CSS Modules) for content management and distribution. Backend service developed in Golang to handle AI processing pipelines, designed for production-grade deployment.

Hidayah Canopy — Business Web App

HTML · CSS · JavaScript

Website for a canopy installation and sales service business, providing product catalog, service information, and customer contact features.

Dumphere — Android Photo Upload App

Android · Java/Kotlin

Mobile application for Android enabling users to upload and capture photos, designed with a focused and intuitive single-purpose interface.

Respiratory Disease Detection — ML Competition

Python · TensorFlow · Deep Learning

Machine learning model for predicting and detecting respiratory diseases through chest X-ray image classification, built for a competitive ML challenge.

SKILLS

Hard Skills:	Python, PHP, Laravel, Flask, Golang (basic), Next.js, HTML, JavaScript, TypeScript, C, C++, MySQL, PostgreSQL, Git/GitHub, Pandas, NumPy, TensorFlow, WebGL, Three.js, Canva, Microsoft Office, Chatbot AI, Photography
Soft Skills:	Leadership · Problem Solving · Analytical Thinking · Adaptability · Time Management · Attention to Detail
Trainings:	LKMW-TD · PKTI-TD · LKMM-PRA TD · LBE Informatics Programming Laboratory
Webinars:	Youth Today with AIESEC (2024)